

Drone Safety System

Hardware Costing & Student Buyer's Guide

*Full bill of materials, component-by-component comparisons,
and budget builds for a trigger-driven autonomous UAV.*

Four ways to build it:

Tier	Cost	What it gets you
Simulation only	₹0	Full software on any PC - all dev happens here
Student / rock-bottom	~₹23,800	Cheapest real quad that flies the stack safely
Minimum airframe	~₹28,400	Proven workhorse build (Pixhawk 2.4.8)
Recommended	~₹51,600	Current-gen autopilot + telemetry + 2nd battery

Prices verified June 2026 against robu.in / Robokits / IndiaMART / ElectroPi listings; they drift over time - treat every figure as plus/minus 15%. All amounts in Indian Rupees (₹, INR).

1. Start here: spend nothing first

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This project is a software-first system. The entire flight stack - auto-arm, take-off, fly-to-GPS, hover, return, land, plus every failsafe - runs in ArduPilot SITL (Software In The Loop) on any laptop for zero rupees. You can dispatch, divert, recall and failsafe-abort hundreds of simulated missions before a single screw is turned.

Honest advice for students

Spend ₹0 first. Run the simulator until you can predict what the aircraft will do before it does it. The hardware below then behaves like a faster, windier, more expensive version of the simulator you already trust. Buy parts only when you have a real reason to fly.

When you are ready for real flight, the rest of this guide gives you (2) four complete budget builds, (3) a part-by-part comparison so you can mix and match by price, (4) where the money actually goes, and (5) student money-saving tactics and traps to avoid.

Safety & legality (India)

Civil drone flight is governed by the Drone Rules, 2021. This class of build is Micro/Small category: register on DigitalSky for a UIN, fly only in green-zone airspace, and ALWAYS keep a trained pilot with an RC transmitter and kill switch in visual line of sight. Autonomous does not mean unsupervised.

2. The four complete builds

2.1 Student / rock-bottom build (~₹23,800)

The cheapest configuration that still flies this stack safely. It swaps the Pixhawk for a low-cost ArduPilot-capable flight controller and assumes you already own a laptop to act as the ground station (no dedicated companion computer yet - you tether over USB/telemetry for early flights).

#	Item	Spec / example	Qty	Price (₹)
1	Frame	F450-class glass-fibre quad + PDB	1	1,200
2	Flight controller	SpeedyBee F405 V3 / Matek F405 (runs ArduPilot 4.x)	1	4,800
3	GPS + compass	u-blox NEO-M8N with stand	1	2,000
4	Motors	2212 920kV brushless	4	2,200
5	ESCs	30A BLHeli, 3S	4	2,000
6	Propellers	1045 self-locking + spares	2 sets	650
7	Battery	3S 2200mAh 30C LiPo, XT60	1	1,200
8	Power / BEC	Voltage+current sense module + 5V BEC	1	600
9	RC TX + RX	FlySky FS-i6 + FS-IA6B (6ch)	1	4,200
10	Charger	IMAX B6-class balance charger + supply	1	2,000
11	Telemetry (optional but advised)	433/915MHz SiK radio pair	1	2,200
12	Consumables	XT60, silicone wire, heat-shrink, zip ties, thread-lock	-	750
Total				₹~23,800

- Trade-off: an F405 board has less RAM/flash and fewer redundant IMUs than a Pixhawk. It runs the same ArduPilot firmware and the same parameter set, so this codebase does not change - but you lose dual-IMU redundancy. Fine for line-of-sight student flights; step up before BVLOS.
- Smaller 2200mAh pack = ~6-8 min flight. Cheap and light for learning; buy a 5200mAh pack once you stop crashing.
- No companion computer: run the API on your laptop tethered to the FC. Add a Pi later (+₹2,800) for on-board autonomy.

2.2 Minimum build (~₹28,400)

The proven, recommended-for-most-people starting point from the project's Build & Operations Guide: a real Pixhawk and an on-board Raspberry Pi so the aircraft is autonomous without a tethered laptop.

#	Item	Spec / example	Qty	Price (₹)
1	Frame	F450-class glass-fibre quad + PDB	1	1,200
2	Flight controller	Pixhawk 2.4.8 (FMUV2/v3) + buzzer + safety switch	1	6,500
3	GPS + compass	u-blox NEO-M8N with stand	1	2,000
4	Motors	2212 920kV brushless	4	2,400
5	ESCs	30A BLHeli, 3S	4	2,200
6	Propellers	1045 self-locking + spares	2 sets	700
7	Battery	3S 5200mAh 35C LiPo, XT60	1	3,000

#	Item	Spec / example	Qty	Price (₹)
8	Power module	Pixhawk power module (V+I sense, XT60)	1	500
9	RC TX + RX	FlySky FS-i6 + FS-IA6B (6ch)	1	4,200
10	Companion computer	Raspberry Pi Zero 2 W + 32GB card + UART cable	1	2,800
11	Charger	IMAX B6-class balance charger + supply	1	2,000
12	Consumables	XT60, silicone wire, heat-shrink, zip ties, thread-lock	-	900
	Total			₹~28,400

2.3 Recommended build (~₹51,600)

Everything in the Minimum build, with these upgrades for current-gen reliability and sustained testing throughput:

Upgrade	Why	Delta (₹)
Pixhawk 6C / Cube Orange Lite (vs 2.4.8)	Current-gen IMUs, better EKF, vibration isolation	+10,500
M9N / M10 GPS (vs M8N)	Faster fix, better multipath rejection	+2,000
Raspberry Pi 4 (4GB) (vs Zero 2 W)	Headroom for camera + on-board dashboard	+3,000
Second 5200mAh battery	Continuous test cycles, double throughput	+3,000
433/915MHz SiK telemetry pair	Live telemetry + params from the bench, no USB	+3,500
Spare props x4 sets + spare motor	You WILL break props learning	+1,200
Subtotal of upgrades over Minimum		+23,200

3. Part-by-part comparison (mix & match by price)

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Each subsystem below lists a Budget / Mid / Better option. Build your own tier by picking one row per subsystem. The 'Pick for students' note tells you the value choice.

3.1 Flight controller (the single biggest cost lever)

Option	Key specs	Notes	Price
SpeedyBee/Matek F405	F405 MCU, 1 IMU, baro, 5 UART	Runs ArduPilot 4.x; no IMU redundancy	4,000-5,300
Pixhawk 2.4.8 (clone)	FMUv2/v3, dual IMU, 5 UART	Budget workhorse; same params as costly boards	5,000-7,000
Pixhawk 6C	STM32H7, triple IMU, vibration-iso	Current-gen, far better EKF behaviour	16,000-19,000
Cube Orange Lite/Plus	H7, isolated IMU, robust	Pro-grade; overkill for first build	24,000-32,000

Pick for students: The Pixhawk 2.4.8 has dropped to ~₹6,500 - only ~₹1,700 over the F405 - so take it for dual-IMU redundancy unless you need absolute rock-bottom, then the F405.

3.2 GPS + compass module

Option	Key specs	Notes	Price
NEO-6M	~1Hz, older, no good compass	AVOID for autonomy - slow, weak fix	600-1,000
NEO-M8N	10Hz, GPS+GLONASS, HMC/IST compass	The sweet spot; what the project uses	1,100-2,200
NEO-M9N / M10	Multi-band, faster fix, better multipath	Worth it in cluttered/urban sites	3,800-4,800

Pick for students: NEO-M8N. The M9N is a real upgrade only near buildings; skip the M6N entirely.

3.3 Frame

Option	Key specs	Notes	Price
F450 clone (glass-fibre)	450mm, integrated PDB	Cheap, repairable, plenty of room	900-1,400
S500 (glass-fibre)	500mm, more payload room	Steadier, fits bigger props/battery	1,600-2,400
Carbon-fibre 450/500	Stiffer, lighter, pricier	Nice-to-have, not needed for this mission	3,500-6,000

Pick for students: F450 glass-fibre. Cheapest, easiest to repair after the inevitable learning crash.

3.4 Motors + ESCs (propulsion, buy as a matched set of 4)

Option	Key specs	Notes	Price (set of 4)
2212 920kV + 30A ESC	3S, 1045 props, ~800g thrust/motor	Standard F450 combo; well documented	4,000-5,000
2213 935kV + 30A BLHeli-S	Slightly more thrust, smoother	Good if you add weight (Pi, telemetry)	5,000-6,500
4-in-1 ESC + 2306 motors	Compact, clean wiring	More for racing; not needed here	6,500-9,000

Pick for students: 2212 920kV + 30A ESC. Cheapest proven combo; spares are everywhere.

3.5 Battery (3S LiPo) - your main recurring cost

Option	Key specs	Notes	Price
3S 2200mAh 30C	~6-8 min, light	Great for learning; cheap to replace if puffed	900-1,500
3S 5200mAh 35C	~10-12 min	Project default; best flight-time value	2,600-3,500
3S 6000mAh 50C	~13-15 min	Heavier; diminishing returns on small quad	4,500-6,000

Pick for students: Start with a cheap 2200mAh to learn on, then add a 5200mAh. Buying two small packs beats one big one for test throughput.

3.6 RC transmitter + receiver (safety-critical: do not cheap out blindly)

Option	Key specs	Notes	Price
FlySky FS-i6 + iA6B	6ch, iBUS/PPM, 2.4GHz	Reliable budget standard; kill switch ok	4,000-5,000
RadioMaster Pocket + RX	ELRS, expandable, model memory	Future-proof; long range	5,500-8,000
FrSky Taranis-class	16ch, telemetry, premium feel	More than this project needs	12,000+

Pick for students: FlySky FS-i6. Cheap, dependable, and supports the mandatory pilot override + kill switch. Never run a build without a working RC override.

3.7 Companion computer (runs flight_core + the API on-board)

Option	Key specs	Notes	Price
None (laptop tether)	Use your own laptop as GCS	Free; fine for early line-of-sight flights	0
Raspberry Pi Zero 2 W	Quad A53, 512MB	Runs the stack; project default (board ~1,900)	1,900-3,200
Raspberry Pi 4 (2-4GB)	Quad A72, 2-4GB	Headroom for dashboard + camera later	5,000-8,000

Pick for students: Skip it at first (tether your laptop), then a Pi Zero 2 W when you want the aircraft to be self-contained.

3.8 Charger + accessories

Option	Key specs	Notes	Price
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Option	Key specs	Notes	Price
IMAX B6 clone + 12V supply	Balance, 1-6S, 50W	Essential; never charge LiPo without balance	1,500-2,200
ISDT / HOTA dual charger	Faster, safer, 2 packs	Nice once you own several batteries	4,500-7,000
LiPo safe bag + checker	Storage + cell-voltage alarm	Cheap insurance against fire	400-700

Pick for students: IMAX B6-class + a ₹150 cell checker + a LiPo safe bag. Non-negotiable safety basics.

4. Where the money actually goes

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On the Minimum build, three line items - the flight controller, the RC set, and the battery - are roughly half the cost. Those are exactly the items you should NOT cheap out on for safety, except the FC where a credible cheaper option (F405) exists.

Item	Cost (₹)	Share
Flight controller	6,500	23%
RC transmitter + receiver	4,200	15%
Battery	3,000	11%
Companion computer	2,800	10%
Motors	2,400	8%
ESCs	2,200	8%
Charger	2,000	7%
GPS + compass	2,000	7%
Frame	1,200	4%
Consumables	900	3%
Props	700	2%
Power module	500	2%
Total (Minimum build)	28,400	100%

What changed since the original guide

Pixhawk 2.4.8 (₹10,500 -> ~₹6,500) and the NEO-M8N GPS (₹2,500 -> ~₹2,000) have fallen sharply, so the Minimum build dropped from ~₹35,600 to ~₹28,400. The Pixhawk is now so cheap that the F405 saves only ~₹1,700 - the Student build's real savings now come from dropping the on-board Pi (tether a laptop, -₹2,800) and starting on a small 2200mAh pack (-₹1,800).

5. Student money-saving tactics

- Simulate first, buy later. Every rupee of hardware is optional until you have flown the SITL stack dozens of times.
- Share a kit. One FC + RC + charger can be a lab/club asset; students buy their own consumables (props, a battery).
- Buy the airframe and a cheap 2200mAh pack to learn on. Crash damage is cheaper when the parts are cheap.
- Reuse a laptop as the ground station - defer the Raspberry Pi until you actually need on-board autonomy.
- Buy props in bulk (you will break them) and motors as a matched set of 4 to keep thrust balanced.
- Watch for festival/clearance sales on robu.in, robokits, quartzcomponents; FC and RC prices swing the most.
- Skip what the mission does not use: no camera/gimbal, no lidar/optical-flow, no 4G link for the first build.

Traps to avoid

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- NEO-6M GPS - too slow/weak for reliable autonomy. The few hundred rupees saved is not worth it.
- No-name ESCs without BLHeli - random behaviour during arming. Buy known brands.
- Charging LiPo without a balance charger or safe bag - a genuine fire risk, not a corner to cut.
- Cheapest possible RC - the transmitter is your pilot override and kill switch; keep it dependable.
- Setting ARMING_CHECK=0 on real hardware to 'make it arm' - never do this off the simulator.

Bottom line for a student team

Target the ~₹23,800 Student build, or ~₹27,000 if you add an on-board Pi and telemetry radio across the team. Spend ₹0 in simulation until you are confident, buy a cheap battery to learn on, and only upgrade the flight controller and battery once you have stopped crashing.

Generated from the project's Build & Operations Guide; prices refreshed against robu.in / Robokits / IndiaMART / ElectroPi listings, June 2026, plus/minus 15% - verify before buying. Companion documents: BUILD_AND_OPERATIONS_GUIDE.md, HARDWARE_INTEGRATION.md.